WORKED SOLUTION

STEP III 2017, Question 11

Mechanics • Gun carriage

STEP III 2017, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q11

This question has a full guided treatment in *Beyond STEP*, Section 10.3, *Rockets and Guns*: https://beyondstep.keithmcnulty.org/ch10/rockets.html

The key idea

No external horizontal force acts on the system, so horizontal momentum is conserved at every firing. The odd-looking ground speed $v - V$ is really saying something simple: the projectile always leaves at speed v *relative to the truck*. For the kinetic energy, don't recompute everything from scratch each time; use $(*)$ to simplify the change in energy at one firing, and then add up.

Solution

Model the truck, guns and projectiles as particles moving along one line, and take the forward direction (the way the truck moves) as positive. A projectile fired when the truck has speed V has velocity $-(v - V) = V - v$, that is, velocity $-v$ relative to the truck. The firings are internal to the system and the track is smooth, so the total horizontal momentum is conserved at each firing.

Part (i)

Before firing everything is at rest, so the momentum is zero. With $V = 0$, each projectile leaves with velocity $-v$, and the truck (mass M) moves forwards with speed u:

$$Mu - nmv = 0 \quad \Longrightarrow \quad u = \frac{nmv}{M}.$$

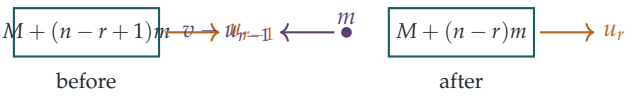
The system started with no kinetic energy, so the energy gained is

$$K = \tfrac{1}{2}Mu^2 + \tfrac{1}{2}nmv^2 = \tfrac{1}{2}\,\frac{n^2m^2v^2}{M} + \tfrac{1}{2}nmv^2 = \tfrac{1}{2}nmv^2\left(1 + \frac{nm}{M}\right),$$

as required.

Part (ii)

Consider the rth firing. Just before it, the truck carries $n-r+1$ unfired projectiles, so the moving mass is $M+(n-r+1)m$ with speed u_{r-1}. Just afterwards, mass $M+(n-r)m$ moves with speed u_r and one projectile of mass m has velocity $u_{r-1}-v$. The projectiles fired earlier are unaffected and can be left out.



Conservation of momentum gives

$$\big(M+(n-r+1)m\big)u_{r-1} = \big(M+(n-r)m\big)u_r + m(u_{r-1}-v).$$

Write the left-hand side as $\big(M+(n-r)m\big)u_{r-1} + mu_{r-1}$. The mu_{r-1} terms then cancel, leaving

$$\big(M+(n-r)m\big)(u_r-u_{r-1}) = mv, \qquad \text{so} \qquad u_r - u_{r-1} = \frac{mv}{M+(n-r)m},$$

which is $(*)$.

Adding $(*)$ for $r=1,\dots,n$, the left-hand side telescopes and $u_0=0$, so

$$u_n = \sum_{r=1}^{n} \frac{mv}{M+(n-r)m}.$$

Each denominator satisfies $M+(n-r)m \geq M$, so each term is at most $\dfrac{mv}{M}$. The inequality is strict for $r=1$, because $n>1$ makes $(n-1)m>0$. Hence

$$u_n < n\cdot\frac{mv}{M} = u.$$

Ask yourself Why should firing one at a time be worse?

In the early firings the gun has to push not only the truck but also all the projectiles still on board, so each shot gives a smaller increase in speed. The term $r=n$ (the last shot, with nothing left on board) is the only one that matches a share of the simultaneous firing.

Part (iii)

At the rth firing, only the truck (with its remaining load) and the projectile just fired change their velocities. So

$$\begin{aligned} K_r - K_{r-1} &= \tfrac{1}{2}\big(M+(n-r)m\big)u_r^2 + \tfrac{1}{2}m(u_{r-1}-v)^2 - \tfrac{1}{2}\big(M+(n-r+1)m\big)u_{r-1}^2 \\ &= \tfrac{1}{2}\big(M+(n-r)m\big)(u_r^2-u_{r-1}^2) + \tfrac{1}{2}m\big[(v-u_{r-1})^2 - u_{r-1}^2\big] \\ &= \tfrac{1}{2}\big(M+(n-r)m\big)(u_r-u_{r-1})(u_r+u_{r-1}) + \tfrac{1}{2}m(v^2-2vu_{r-1}). \end{aligned}$$

By $(*)$, $\big(M+(n-r)m\big)(u_r-u_{r-1})=mv$, so the first term is $\frac{1}{2}mv(u_r+u_{r-1})$. Therefore

$$\begin{aligned}K_r-K_{r-1}&=\tfrac{1}{2}mvu_r+\tfrac{1}{2}mvu_{r-1}+\tfrac{1}{2}mv^2-mvu_{r-1}\\&=\tfrac{1}{2}mv^2+\tfrac{1}{2}mv(u_r-u_{r-1}),\end{aligned}$$

as required.

Adding this for $r=1,\dots,n$, and using $K_0=0$ and $u_0=0$, both sides telescope:

$$K_n=\tfrac{1}{2}nmv^2+\tfrac{1}{2}mv(u_n-u_0)=\tfrac{1}{2}nmv^2+\tfrac{1}{2}mvu_n.$$

Finally, compare with part (i). Since $u=nmv/M$,

$$K=\tfrac{1}{2}nmv^2+\tfrac{1}{2}\,\frac{n^2m^2v^2}{M}=\tfrac{1}{2}nmv^2+\tfrac{1}{2}nmv\,u.$$

From part (ii), $u_n<u$, and $u<nu$ because $u>0$ and $n>1$. As $mv>0$,

$$K_n=\tfrac{1}{2}nmv^2+\tfrac{1}{2}mvu_n<\tfrac{1}{2}nmv^2+\tfrac{1}{2}mvu<\tfrac{1}{2}nmv^2+\tfrac{1}{2}nmvu=K.$$

So $K_n<K$: firing one at a time releases less kinetic energy.

Checks

- Take $n=3$, $M=10$, $m=1$, $v=1$. Then $u=0.3$, while $(*)$ gives $u_3=\frac{1}{12}+\frac{1}{11}+\frac{1}{10}=\frac{181}{660}\approx 0.274$, which is smaller, as it should be. The energies are $K=1.95$ and $K_3=\frac{2161}{1320}\approx 1.637$.
- Working out the momentum and the kinetic energy of every body directly after each shot, for thousands of random choices of n, M, m and v, agrees exactly with $(*)$ and with the formula for K_n; this was confirmed by computer.
- Limiting case: if $m\ll M$, every term of $(*)$ is close to mv/M, so $u_n\approx u$, and both energies are close to $\frac{1}{2}nmv^2$. When the truck is very heavy, the order of firing hardly matters.

Common slips

- Forgetting that the unfired projectiles ride with the truck, and using mass M instead of $M+(n-r+1)m$ before the rth shot.
- Giving the projectile velocity $-v$ instead of $u_{r-1}-v$: the question fixes its speed relative to the ground as $v-V$, not v.
- Including the kinetic energy of earlier projectiles in K_r-K_{r-1}. They are unaffected and cancel.

Beyond STEP

This is a discrete version of rocket motion. If the n shots are replaced by a continuous stream of exhaust, the sum for u_n becomes an integral and gives the Tsiolkovsky rocket equation, $\Delta u = v \ln(m_0/m_1)$. A rocket is in the position of the truck firing one shot at a time: its fuel can't all be burnt in a single instant, so it pays the price that part (ii) uncovers. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats variable-mass systems and rockets carefully.

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